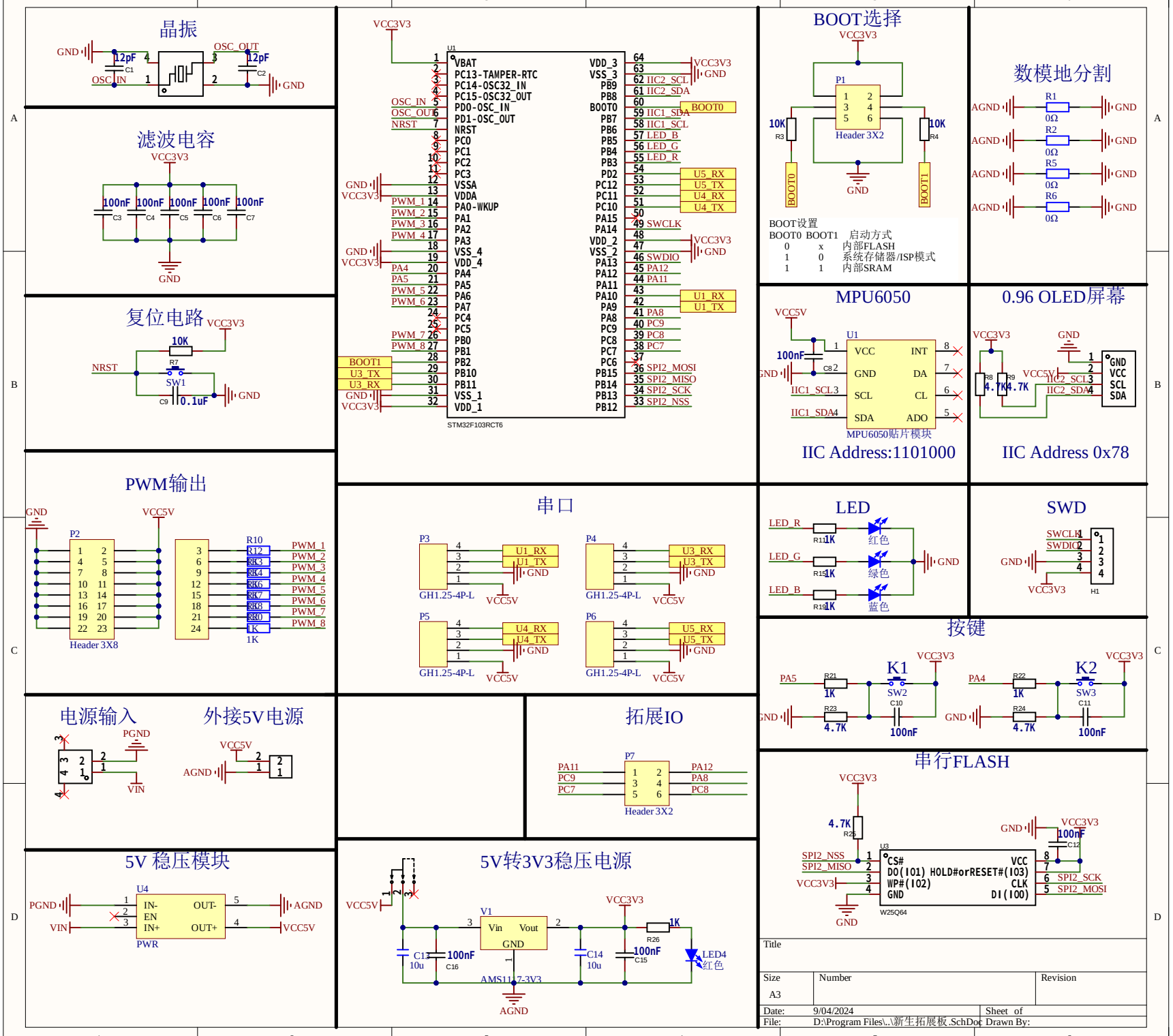
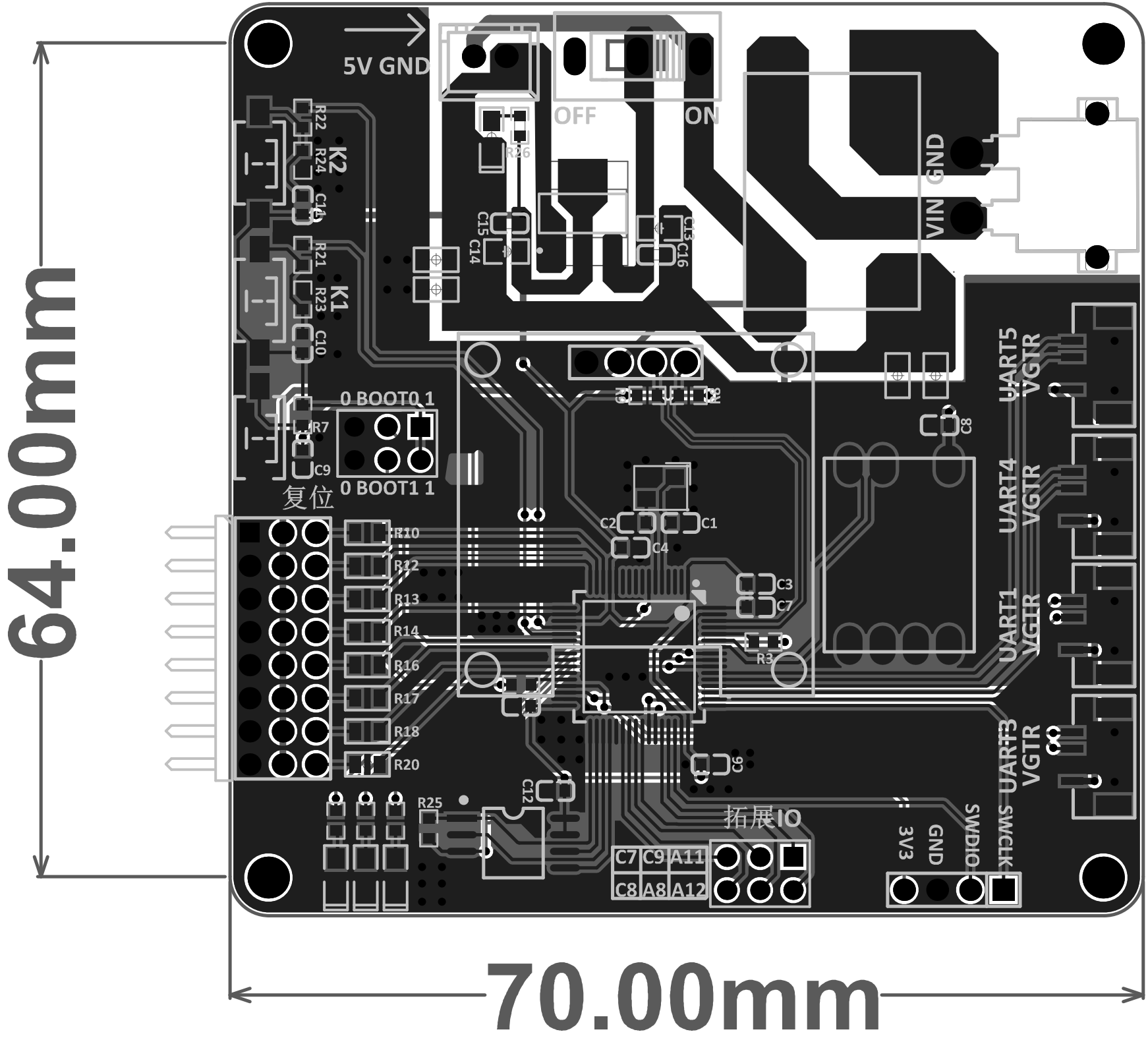

晶振
GND
12pF
C1
OSC_IN
OSC_OUT
12pF
C2
GND
滤波电容
VCC3V3
100nF C3
100nF C4
100nF C5
100nF C6
100nF C7
GND
复位电路
VCC3V3
10K
R7
NRST
SW1
C9 0.1uF
GND
PWM输出
GND
VCC5V
P2
Header 3X8
R10
1K
PWM_1
PWM_2
PWM_3
PWM_4
PWM_5
PWM_6
PWM_7
PWM_8
电源输入
PGND
VIN
外接5V电源
VCC5V
AGND
5V 稳压模块
U4
PGND
VIN
IN-
EN
IN+
OUT-
OUT+
AGND
VCC5V
PWR
VCC3V3
U1
STM32F103RCT6
VBAT
PC13-TAMPER-RTC
PC14-OSC32_IN
PC15-OSC32_OUT
PD0-OSC_IN
PD1-OSC_OUT
NRST
PC0
PC1
PC2
PC3
VSSA
VDDA
PA0-WKUP
PA1
PA2
PA3
VSS_4
VDD_4
PA4
PA5
PA6
PA7
PC4
PC5
PB0
PB1
PB2
PB10
PB11
VSS_1
VDD_1
VDD_3
VSS_3
PB9
PB8
BOOT0
PB7
PB6
PB5
PB4
PB3
PD2
PC12
PC11
PC10
PA15
PA14
VDD_2
VSS_2
PA13
PA12
PA11
PA10
PA9
PA8
PC9
PC8
PC7
PC6
PB15
PB14
PB13
PB12
OSC_IN
OSC_OUT
NRST
BOOT1
U3_TX
U3_RX
PWM_1
PWM_2
PWM_3
PWM_4
PWM_5
PWM_6
PWM_7
PWM_8
IIC2_SCL
IIC2_SDA
IIC1_SCL
IIC1_SDA
LED_B
LED_G
LED_R
SWCLK
SWDIO
U5_RX
U5_TX
U4_RX
U4_TX
U1_RX
U1_TX
PA12
PA11
PA8
PC9
PC8
PC7
SPI2_MOSI
SPI2_MISO
SPI2_SCK
SPI2_NSS
串口
P3
U1_RX
U1_TX
GND
GH1.25-4P-L
VCC5V
P4
U3_RX
U3_TX
P5
U4_RX
U4_TX
P6
U5_RX
U5_TX
拓展IO
P7
Header 3X2
PA11
PC9
PC7
PA12
PA8
PC8
5V转3V3稳压电源
VCC5V
V1
Vin
Vout
GND
AMS1117-3V3
C13 10u
C16 100nF
C14 10u
C15 100nF
R26 1K
LED4
红色
AGND
BOOT选择
VCC3V3
P1
Header 3X2
10K
R3
10K
R4
BOOT0
BOOT1
GND
BOOT设置
BOOT0 BOOT1 启动方式
0 x 内部FLASH
1 0 系统存储器/ISP模式
1 1 内部SRAM
数模地分割
AGND
R1 0Ω
R2 0Ω
R5 0Ω
R6 0Ω
GND
MPU6050
VCC5V
U1
VCC
GND
SCL
SDA
INT
DA
CL
ADO
100nF
C8
IIC1_SCL
IIC1_SDA
MPU6050贴片模块
IIC Address:1101000
0.96 OLED屏幕
VCC3V3
GND
R8 4.7K
R9 4.7K
VCC5V
IIC2_SCL
IIC2_SDA
GND
VCC
SCL
SDA
IIC Address 0x78
LED
LED_R
LED_G
LED_B
R11 1K
R15 1K
R19 1K
红色
绿色
蓝色
GND
SWD
SWCLK
SWDIO
GND
VCC3V3
H1
按键
K1
K2
SW2
SW3
PA5
PA4
R21 1K
R22 1K
R23 4.7K
R24 4.7K
C10 100nF
C11 100nF
VCC3V3
GND
串行FLASH
VCC3V3
4.7K R25
U3
CS#
DO(IO1) HOLD#orRESET#(IO3)
WP#(IO2)
GND
VCC
CLK
DI(IO0)
W25Q64
SPI2_NSS
SPI2_MISO
SPI2_SCK
SPI2_MOSI
C12 100nF
Title
Size A3
Number
Revision
Date: 9/04/2024
Sheet of
File: D:\Program Files\..\新生拓展板.SchDoc
Drawn By:


64.00mm
70.00mm
5V GND
OFF
ON
VIN
GND
K2
K1
0 BOOT0 1
0 BOOT1 1
复位
UART5
UART4
UART1
UART3
VGTR
拓展IO
SWDIO
SWCLK
3V3
GND

Comment	Designator	Footprint	LibRef	Quantity
12pF	C1, C2	C0603	C0603	2
100nF	C3, C4, C5, C6, C7, C8, C10, C11, C12, C15, C16	C0603	C0603	11
0.1uF	C9	C0603	C0603	1
10u	C13, C14	C 0805_L	C	2
XT30PW-M	CN1	CONN-TH_XT30PW-M	XT30PW-M	1
XH-2A	CN2	CONN-TH_XH-2A	XH-2A_C20079	1
PZ254V-11-04P	H1	HDR-TH_4P-P2.54-V-M	HDR-2.54-4P	1
红色	LED1, LED4	LED 0805G	LED-SMD	2
绿色	LED2	LED 0805G	LED-SMD	1
蓝色	LED3	LED 0805G	LED-SMD	1
Header 3X2	P1, P7	HDR2X3	Header 3X2	2
Header 3X8	P2	Header 3X8	Header 3X8	1
GH1.25-4P-L	P3, P4, P5, P6	GH1.25-4P-L	GH1.25-4P-L	4
0Ω	R1, R2, R5, R6	R 0805_L	R	4
10K	R3, R4, R7	R0603	R0603	3
4.7K	R8, R9, R23, R24, R25	R0603	R0603	5
1K	R10, R12, R13, R14, R16, R17, R18, R20	R 0805_L	R	8
1K	R11, R15, R19, R21, R22, R26	R0603	R0603	6
3x6x5	SW1, SW2, SW3	TSW SMD-3*6*2.5	TSW 3x6	3
HX SS12D10G4	SW4	SW-TH_SHOU-HAN_SS12D10G4 - duplicate	Switch-3	1
MPU6050贴片模块	U1	MPU6050	MPU6050贴片模块	1
STM32F103RCT6	U1	LQFP-64_L10.0-W10.0-P0.50-LS12.0-BL	STM32F103RCT6	1
HS96L03W2C03	U2	OLED-TH_L27.8-W27.2-P2.54_C9900033791	HS96L03W2C03	1
W25Q64	U3	SOP8W_N	W25Q64JVSSIQ	1
PWR	U4	PWR	PWR	1
AMS1117-3V3	V1	SOT223_L	AMS1117	1
8MHz	X1	CRYSTAL-SMD_4P-L3.2-W2.5-BL	3225 8M 12PF 10PPM	1